

REMORIN-MEDIATED PROTEIN-LIPID NETWORKING DURING NANODOMAIN FORMATION IN PLANT CELLS

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ABSTRACT

Biological membranes are highly dynamic platforms that can regulate essential cellular processes through the recruitment and organization of peripheral membrane proteins into functional supramolecular assemblies. In plants, remorins are multifunctional proteins, regulating immunity, development and symbiosis (1). Their biological functions depend on membrane association, where they sequester specific lipids into functional membrane nanodomains. By integrating solution and solid-state NMR spectroscopy, advanced molecular imaging, and molecular dynamics simulations, we uncover the structural principles that govern remorin membrane recognition, self-assembly, and biological function (2, 3, 4). We show that remorins form coiled-coil oligomeric pre-assemblies decorated with a dynamic, fuzzy region and a barcode-like distribution of positively charged residues prior to membrane binding. Membrane association then promotes lipid sequestration, modulates membrane biophysical properties, and reorganizes the electrostatic landscape at the protein–membrane interface (5). Together, our findings provide a molecular framework for understanding how remorins couple protein self-assembly with lipid organization to generate functional membrane nanodomains.

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